

Discoverable Transport: Closure, Observability, and the Cost of Acceleration

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Abstract—An iteration may expose a simple direction without exposing a simple law for its future motion. Discoverable transport concerns the stronger task of acquiring a representation that can be advanced more cheaply than the ordinary computation it replaces. We develop this question for mirror, Bregman, and entropic iterations through three distinct requirements: closure on the relevant states or requests, acquisition of the information needed by that closure, and economical evaluation with controlled error. Finite invariant observables permit exact powering; a trajectory-defect identity separates compression from evolution of the local geometry. For positive-kernel Sinkhorn iteration, conditional variance bounds the nonlinear forcing, and a finite energy budget yields horizon-uniform first- and second-jet error bounds at a fixed point, without inverse spectral gap factors. The constants retain their amplitude and marginal dependence. Exact linear observability and finite joint-pattern algebras give scoped limits on the dimension required to carry a rule or a primitive response. These results explain why slow motion, low snapshot rank, and exact algebraic identities can each coexist with unsuccessful acceleration. Selected experiments illustrate certified affine-piece discovery, supplied projective closure, size-dependent primitive reuse, and loss of amortization through curvature or request-family growth. Primitive reuse improves four of six reported 1024-point trajectories, but loses on the narrowest kernels and at smaller size. The resulting theory identifies sufficient structures and representation-specific obstructions; it does not assert a universal accelerator or a universal lower bound on acceleration.

Index Terms—Iterative methods, mirror descent, Bregman iteration, Sinkhorn scaling, Krylov subspaces, observable closure, conditional variance, model reduction.

I. INTRODUCTION

A slowly changing iterate often suggests that many future steps should be replaceable by one calculation. The suggestion is incomplete. The direction may rotate, its rate may depend on unobserved variables, or the representation needed to carry it may cost more than the steps it replaces. The relevant question is therefore not only whether motion is predictable, but which relations make that prediction both valid and economical.

We call this problem *discoverable transport*. Its object is the complete state of an existing iteration, or the requests made to an expensive primitive inside that iteration. The representation is acquired from available expressions, operator actions, and observed states; its validity concerns the original map. A method that instead reaches the same minimizer by a different trajectory addresses a related acceleration problem, but does not thereby establish finite-trajectory transport.

Three distinctions organize the construction. *Closure* determines whether the retained variables predict what is required next. *Observability and acquisition* determine whether those variables and relations can be obtained from the permitted

information. *Amortization* determines whether their entry, composition, validation, and readout costs are repaid before the representation ceases to be useful. These are requirements of the transport constructions studied here, not a classification theorem for every possible acceleration algorithm.

The distinction is visible already in mirror geometry. A dual Jacobian is self-adjoint in a natural metric, making finite Krylov transport available. Yet increasing its dimension removes projection error without removing the error of freezing the geometry. Nonlinear observable coordinates can instead close the actual dynamics. Their existence depends on the map and the representation, not on curvature alone. In entropic scaling, the conditional geometry evolves exactly within a fixed diagonal scaling orbit; carrying that orbit still need not be cheaper than executing an ordinary pass.

The main results sharpen these observations. We state the exact observable and approximate path contracts, derive conditional-energy bounds for Sinkhorn, and separate two finite-dimensional obstructions: the outputs needed to carry an evolving rule may require more directions than the state, and pointwise nonlinear requests may generate a much larger algebra than the linear span of their log directions. These obstructions are explicitly relative to a representation class. Supplied block and Gaussian families, certified affine pieces, and measured primitive responses provide positive counterpoints. The experiments serve to distinguish the theoretical conditions rather than to enumerate successive implementations.

II. RELATION TO EXISTING ACCELERATION

Krylov acceleration is established numerical analysis. Anderson acceleration on linear problems is related to GMRES [1], nonlinear extrapolation uses histories of iterates [2], and GMRES acceleration of quadratic ADMM has been analyzed [3]. The finite-flow polynomial below is a classical matrix-function construction [4]; on a fixed quadratic its stationary limit is a conjugate-gradient Galerkin correction [5]. Neither its polynomial identity nor the application of Krylov methods to Bregman-related iterations is claimed as new.

Poon and Liang already fit finite companion recurrences to displacement histories and sum their powers over a finite forecast horizon [6]. Their earlier analysis accounts for fitting work and multistep residual error [7]. Mai and Johansson accelerate proximal-gradient iterations in pre-proximal coordinates [8]; geometric Bregman acceleration is also a separate established program [9]. The distinction here is between an acquired recurrence, its departure from the original finite trajectory, and its complete execution cost.

Finite invariant observable spaces belong to Koopman theory [10], [11]. QuACK already alternates ordinary optimizer

steps with learned forecasts and loss-based acceptance [12]. Invariance diagnostics and multistep Koopman error bounds are studied directly in [13]. Hidden-variable memory and delay representations have related formulations in Koopman–Mori–Zwanzig reduction [14], [15]. The observable framework used below is thus a common foundation, not a new name for an undiscovered linearization principle.

For entropic transport [16], spectral overrelaxation [17] and Arnoldi matrix balancing [18] are close algorithmic comparators. Gaussian Sinkhorn already has finite-dimensional Kalman/Riccati formulations [19], and Riccati products admit structured representations [20]. Our Gaussian example is an instance of this prior structure. The present contribution is the connection between geometry-specific defect and energy bounds, the information required by state-and-rule representations, and the algebra of primitive requests. A complete comparison with all such accelerators, specialized transport solvers, or continuation schedules is not supplied.

III. CLOSED REPRESENTATIONS OF AN ITERATION

Let $F : \Omega \rightarrow \Omega$ be a deterministic map of the complete state. Independent splitting memories belong to that state. A quotient is permitted when it retains everything needed for the specified future output; deletion of a memory variable without such a property changes the map.

A. Finite observables and the changing derivative

Proposition 1 (Finite observable transport). *Suppose $\Phi : \Omega \rightarrow \mathbb{R}^r$ and χ satisfy*

$$\Phi(F(z)) = T\Phi(z), \quad \chi(\Phi(z)) = z$$

on an invariant set. Then, for every nonnegative integer H ,

$$F^H(z) = \chi(T^H\Phi(z)). \quad (1)$$

If the identities hold on an open set and the maps are differentiable,

$$D(F^H)(z) = D\chi(T^H\Phi(z))T^HD\Phi(z). \quad (2)$$

For a smooth invariant manifold the derivative identity applies to its tangent directions; other perturbations require an entry-map analysis.

Proof. Induction gives $\Phi(F^H(z)) = T^H\Phi(z)$; decode and differentiate. $\square$

A fixed lifted operator can therefore carry a changing physical derivative. No explicit matrix of that derivative need be retained. The cyclic minimal polynomial of T reduces its powers; binary composition evaluates them without visiting every intermediate state. Entry, decoding, conditioning, and the arithmetic cost of those powers remain separate obligations. A finite-dimensional nonlinear parameterization alone does not imply fixed-size linear powering. In particular a general polynomial rule may generate successively higher degrees and require an infinite lift.

For example, $u^+ = \lambda u$, $v^+ = \mu v + \beta u^2$ closes on (u, v, u^2) . At the resonance $\mu = \lambda^2$,

$$v_H = \mu^H v_0 + H\mu^{H-1}\beta u_0^2.$$

The resonant interaction is carried by a Jordan block rather than removed by division by $\mu - \lambda^2$. This special triangular closure does not extend to arbitrary polynomial maps. The following algebraic condition identifies a broader sufficient case without assuming that its coordinates can be acquired cheaply.

Proposition 2 (Positive weighted-degree closure). *Let a polynomial map $N : \mathbb{R}^d \rightarrow \mathbb{R}^d$ have positive coordinate weights $w_i > 0$, with every monomial in N_i of weighted degree w_i . For any finite W , monomials of weighted degree at most W span a finite space invariant under composition with N . If $W \geq \max_i w_i$, that space contains all state coordinates and gives a finite linear lift.*

Proof. A monomial x^α has degree $\sum_i \alpha_i w_i$. Substitution of the weighted-homogeneous N_i preserves that degree term by term. There are only finitely many nonnegative integer exponents below W because every w_i is positive. The coordinate monomials are included by the final hypothesis. $\square$

For a supplied Poincaré–Dulac polynomial containing only resonant monomials and with eigenvalues $0 < \lambda_i < 1$, the weights $w_i = -\log \lambda_i$ have this property: a resonant monomial in coordinate k satisfies $\prod_i \lambda_i^{\alpha_i} = \lambda_k$. This statement assumes the normal-form coordinates and an applicable local invariant domain. It proves neither a cheaply discoverable conjugating chart nor a small lift; the number of admissible monomials can be large. Zero or unit eigenvalues are outside this positive-finite-weight argument.

B. Compatibility with a prescribed mirror

Let $\Omega \subset \mathbb{R}^n$ be open and simply connected, let $h \in C^2(\Omega)$ have positive-definite Hessian, and assume ∇h is injective. Let $F : \Omega \rightarrow \Omega$ be C^1 and $\eta > 0$. A mirror step for a differentiable objective satisfies

$$\nabla h(F(x)) = \nabla h(x) - \eta \nabla f(x). \quad (3)$$

Theorem 3 (Geometry–dynamics compatibility). *There exists a C^2 objective f satisfying (3) if and only if $H_h(F(x))DF(x)$ is symmetric everywhere. If Ω is also convex, that objective is convex if and only if*

$$H_h(x) - H_h(F(x))DF(x) \succeq 0 \quad (x \in \Omega). \quad (4)$$

The objective is unique up to an additive constant on a connected domain.

Proof. Define $g(x) = [\nabla h(x) - \nabla h(F(x))]/\eta$. Differentiation gives

$$Dg(x) = \eta^{-1}[H_h(x) - H_h(F(x))DF(x)].$$

Since $H_h(x)$ is symmetric, Dg is symmetric precisely under the stated condition. The closed one-form $g \cdot dx$ is exact on a simply connected domain, giving $g = \nabla f$. The converse follows by differentiating (3). On a convex domain a C^2 function is convex precisely when its Hessian is positive semidefinite. Injectivity of ∇h identifies the prescribed F with the mirror update wherever its image lies in Ω . $\square$

This also tests a candidate finite representation: if ψ is an embedding with decoder χ and $T\psi(\Omega) \subseteq \psi(\Omega)$, set $F = \chi T\psi$ and apply the theorem. Algebraic closure alone does not make the resulting dynamics a gradient method in an arbitrarily assigned mirror.

In dual coordinates $y = \nabla h(x)$, write $G(y) = \nabla h(F(\nabla h^*(y)))$. The equivalent symmetry test is

$$DG(y)H_h(x) = H_h(x)DG(y)^T. \quad (5)$$

For an affine dual map $G(y) = Ay + c$, this must hold for the entire family of mirror Hessians, not just the Hessian at an anchor. For example, $A = \text{diag}(1/2, 4/5)$ passes with $H_h = I$ and fails with $H_h = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$. Conversely, every Legendre geometry admits the family $f = \alpha h - b^T x$, with $G(y) = (1 - \eta\alpha)y + \eta b$ whenever the image stays in the dual domain. Thus arbitrary curvature can coexist with a one-mode affine recurrence. More modes require a compatible Hessian family.

These statements refer to a prescribed mirror algorithm in its affine coordinates. An arbitrary nonlinear coordinate change conjugates an existing map, but recomputing ordinary mirror descent from a transformed objective and transformed potential need not produce that conjugate map.

C. Exact families and the knowledge they require

The simplest family is affine motion. At an anchor z , let $r = F(z) - z$, $J = DF(z)$, and $p_H(t) = \sum_{j=0}^{H-1} t^j$. If F is affine on the visited region, then

$$F^H(z) - z = p_H(J)r. \quad (6)$$

The cyclic space $\mathcal{K}_k(J, r)$ is acquired by operator actions. Exact Arnoldi closure $JQ = QT$ proves exact transport of this orbit for all horizons remaining in that affine region. It counts excited eigenvalues and Jordan structure, rather than ambient dimension. In particular $p_H(1) = H$ carries constant drift even when no stationary dual solution exists.

Affine pieces of a Huber mirror map supply a *discovered* positive example. The current feature constraints and cached tangent features give a first-event horizon and a path certificate (Appendix B). This is stronger evidence than low snapshot rank: the region of validity is checked using the known map expression. For the coupled Meyer recurrence, a faithful quotient retains both vector memories, and disk curvature has an exact pointwise remainder whose fixed linear response can be bounded (Appendix C). An unchanged exterior-disk mask is not an affine piece, so the Huber argument cannot be reused there.

Nonlinear coordinates give a second route. In a scalar mirror with $s = h'(q) > 0$, prescribe

$$s^+ = \frac{\alpha s}{1 + \beta s}, \quad 0 < \alpha \leq 1, \quad \beta \geq 0.$$

The objective derivative $f'(q) = s - \alpha s/(1 + \beta s)$ has

$$f''(q) = \left(1 - \frac{\alpha}{(1 + \beta s)^2}\right) h''(q) \geq 0.$$

Thus, on an invariant positive dual interval, this is convex mirror descent and $(1, 1/s)$ closes linearly. A rational grammar

can identify a fractional-linear relation and synthesize a cross-ratio coordinate from its fixed points. The grammar and mirror factorization are supplied; the fitted relation and its numerical accuracy are learned. Structural verification, not the fit residual alone, establishes exact closure.

Two-group Sinkhorn provides a *supplied-structure* example within a standard algorithm. For a positive 2×2 kernel, the ratio v_1/v_2 obeys a fractional-linear map, hence a 2×2 homogeneous matrix carries both the ratio and its tangent over a finite horizon. A supplied proportional block partition extends the reduction to large supports (Appendix D). Rank two by itself does not identify that partition or certify the reduction.

A continuous Gaussian family provides a different supplied representation. For centered covariance marginals $A, B \succ 0$ and kernel $\exp(-\|x - y\|^2/(2t))$, quadratic scaling potentials remain quadratic. Writing $p \succ 0$ for the conditional precision, a complete pass is

$$p^+ = B^{-1} + (t^2 A^{-1} + p^{-1})^{-1}. \quad (7)$$

Put $T = t^2 A^{-1}$ and $C = B^{-1}$. This is the matrix fractional action of

$$L = \begin{pmatrix} I + CT & C \\ T & I \end{pmatrix}, \quad p^+ = \{(I + CT)p + C\}(Tp + I)^{-1}.$$

Matrix powers compose the recurrence whenever decoding remains valid. The dimension is the physical Gaussian dimension, not a discretized support size. This is established Riccati structure [19], [20]; Appendix E derives the stated convention. Gaussian closure concerns this preserved quadratic-potential family. It does not assert vanishing higher cumulants for arbitrary observables: for a standard Gaussian X , the third cumulant of X^2 is eight.

IV. APPROXIMATE TRANSPORT AND GEOMETRIC DEFECT

Exact closure is a sufficient structure, but a useful finite horizon may survive a controlled departure from it. The following contract makes the required departure explicit before discussing acquisition or speed.

A. A path contract and its cost

For a predicted path $\hat{z}_0 = z$, let $d_j = F(\hat{z}_j) - \hat{z}_{j+1}$. If F is ℓ -Lipschitz between the corresponding actual and predicted points, then

$$\|F^H(z) - \hat{z}_H\| \leq \sum_{j=0}^{H-1} \ell^{H-1-j} \|d_j\|. \quad (8)$$

Indeed the next error is at most ℓ times the current error plus $\|d_j\|$. This elementary recursion is useful only when its defects can be bounded without replaying all the expensive map evaluations.

Let C_F be the cost of one ordinary pass and let

$$C_{\text{tr}} = C_{\text{acquire}} + C_{\text{compose}} + C_{\text{certify}} + C_{\text{read}} + C_{\text{failed}}. \quad (9)$$

Transport at a specified trajectory tolerance saves time only if $C_{\text{tr}} < HC_F$. This is a cost accounting condition, not a theorem that any component can be made small. A convergence

comparison is different. If F is a q -contraction in one fixed norm, $0 < q < 1$, and $\|\hat{z}_H - F^H(z)\| \leq \epsilon \|z - z_*\|$, then

$$\|\hat{z}_H - z_*\| \leq (q^H + \epsilon) \|z - z_*\|.$$

With $c = C_{\text{tr}}/C_F$, the sufficient worst-case advantage is $\epsilon < q^c - q^H$. Local metrics at different anchors cannot be substituted into this comparison without controlling their distortion.

B. Mirror symmetry and frozen finite flow

Let ψ be a Legendre mirror on an open convex primal domain, and assume the required derivatives exist with $\nabla^2\psi(x) \succ 0$. For smooth f , an interior mirror-descent step with fixed $\alpha > 0$ satisfies

$$\nabla\psi(x^+) = \nabla\psi(x) - \alpha\nabla f(x). \quad (10)$$

With $y = \nabla\psi(x)$ and $x = \nabla\psi^*(y)$, its dual map is

$$\mathcal{F}(y) = y - \alpha\nabla f(\nabla\psi^*(y)). \quad (11)$$

All coordinates below are genuine local coordinates. On a probability simplex we use $n - 1$ log ratios; the redundant n -logit gauge would give a singular metric.

Proposition 4 (Local mirror symmetry). *At any interior anchor y , let*

$$M = \nabla^2\psi^*(y) = (\nabla^2\psi(x))^{-1}, \quad B = \nabla^2 f(x).$$

Then

$$J = I - \alpha BM, \quad MJ = M - \alpha MBM = (MJ)^T. \quad (12)$$

Hence J is self-adjoint in $\langle u, v \rangle_M = u^T M v$. Its similarity transform is $M^{1/2} J M^{-1/2} = I - \alpha M^{1/2} B M^{1/2}$. If $\mu \nabla^2\psi(x) \preceq B \preceq L \nabla^2\psi(x)$, its spectrum lies in $[1 - \alpha L, 1 - \alpha \mu]$.

Proof. Differentiate (11). Symmetry follows because both B and M are symmetric. Congruencing the relative Hessian bounds by $M^{1/2}$ gives the spectral statement. No fixed-point condition was used. $\square$

The statement supplies a principled inner product for Arnoldi. In exact arithmetic its reduced matrix is symmetric, so a Lanczos implementation is available. The implementations use twice-reorthogonalized Arnoldi for numerical robustness, charging its full cost. Applying M can itself be expensive; the result is practically useful only when that action is available at an acceptable cost. Diagonal quadratic and separable entropy metrics provide inexpensive actions; general dense mirror metrics need not.

Fix an SPD metric M at the anchor and write $\beta = \|r\|_M$. If $r = 0$, the anchor is unchanged. Otherwise construct an M -orthonormal basis of

$$\mathcal{K}_k(J, r) = \text{span}\{r, Jr, \dots, J^{k-1}r\}.$$

Let $Q^T M Q = I$, $Q e_1 = r/\beta$, and $T_r = Q^T M J Q$. The proposal is

$$\hat{z}_H = z + \delta_{k,H}, \quad \delta_{k,H} = \beta Q p_H(T_r) e_1. \quad (13)$$

Only actions of J are required. The reduced polynomial is evaluated by $c_0 = 0$, $c_{j+1} = \beta e_1 + T_r c_j$, $\delta_{k,H} = Q c_H$. This recurrence does not invert $I - T_r$ and remains well-defined for unit eigenvalues and Jordan blocks. For large horizons, the pair $(T_r^H, p_H(T_r))$ also admits binary composition, so reduced evaluation need not visit every intermediate state.

Let M be fixed during one block, let Q, T_r be the computed projection, and write $E = JQ - QT_r$. For exact Arnoldi $E = h_{k+1,k} q_{k+1} e_k^T$; using the full residual matrix also accounts for floating-point orthogonalization error. Define

$$c_0 = 0, \quad c_{j+1} = \beta e_1 + T_r c_j, \quad v_j = Q c_j, \quad \hat{z}_j = z + v_j.$$

Theorem 5 (Discrete trajectory-defect identity). *If all required evaluations are in the domain of $\mathcal{F}$, define*

$$N(v) = \mathcal{F}(z + v) - \mathcal{F}(z) - Jv.$$

Then

$$d_j := \mathcal{F}(\hat{z}_j) - \hat{z}_{j+1} = N(v_j) + E c_j. \quad (14)$$

If $\mathcal{F}$ is ℓ -Lipschitz in the fixed M -norm between corresponding actual and predicted points, then

$$\|\mathcal{F}^H(z) - \hat{z}_H\|_M \leq \sum_{j=0}^{H-1} \ell^{H-1-j} (\|N(v_j)\|_M + \|E c_j\|_M). \quad (15)$$

If the derivative is L_J -Lipschitz in the induced M -norm on each segment from z to $z + v_j$, then $\|N(v_j)\|_M \leq \frac{L_J}{2} \|v_j\|_M^2$.

Proof. Use $r = \beta Q e_1$ to subtract the reduced recurrence from $\mathcal{F}(z + v_j) = z + r + J v_j + N(v_j)$, giving (14). For $a_j = \mathcal{F}^j(z) - \hat{z}_j$, $a_{j+1} = \mathcal{F}(\mathcal{F}^j(z)) - \mathcal{F}(\hat{z}_j) + d_j$ and $a_0 = 0$. Apply the Lipschitz bound recursively. The derivative-remainder estimate follows by integrating along the segment. $\square$

Increasing the Krylov dimension addresses $E c_j$. It cannot in general remove $N(v_j)$: a larger exact representation of the wrong frozen geometry still predicts the wrong trajectory. At a nonsmooth crossing, (14) remains an algebraic identity for a selected linear action, but the quadratic remainder estimate need not hold. Evaluating every d_j requires the very map calls one hopes to avoid. Direct defect evaluation therefore serves as a separately costed diagnostic; it cannot be treated as a free runtime certificate.

C. Relating the metric error to a Bregman distance

For $x_a = \nabla\psi^*(y_a)$ and $x_b = \nabla\psi^*(y_b)$,

$$D_\psi(x_a, x_b) = D_{\psi^*}(y_b, y_a).$$

If the Hessian along the dual segment is bounded by $aM \preceq \nabla^2\psi^* \preceq bM$, then

$$\frac{a}{2} \|y_b - y_a\|_M^2 \leq D_\psi(x_a, x_b) \leq \frac{b}{2} \|y_b - y_a\|_M^2. \quad (16)$$

This translates the fixed-metric transport error into the original mirror geometry when the metric distortion is controlled. It also identifies a limitation near domain boundaries where the required constants may degenerate.

V. CONDITIONAL ENERGY IN ENTROPIC TRANSPORT

A long relaxation time does not by itself imply large accumulated nonlinear forcing. In Sinkhorn iteration, the same conditional geometry controls the linear dissipation and the curvature. This yields a quantitative distinction between a small spectral gap and failure of an approximate transport model.

Let $K \in \mathbb{R}_{>0}^{m \times n}$ and let a, b be positive marginals of equal total mass. All exponentials, logarithms, and vector quotients below are componentwise. In log column-scaling coordinates, one full pass is

$$x(y) = \log a - \log(Ke^y), \quad F(y) = \log b - \log(K^T e^{x(y)}). \quad (17)$$

The next row scaling is determined by y and carries no independent memory in this ordinary algorithm. This elimination is specific to its phase contract; it does not justify deleting independent splitting memories. The map has the gauge symmetry $F(y + \gamma \mathbf{1}) = F(y) + \gamma \mathbf{1}$.

Set $u = e^{x(y)}$, $v = e^y$, and define

$$\begin{aligned} \Pi &= \text{diag}(u)K \text{diag}(v), & c &= \Pi^T \mathbf{1}, \\ P &= \text{diag}(a)^{-1}\Pi, & Q &= \text{diag}(c)^{-1}\Pi^T. \end{aligned} \quad (18)$$

Thus $\Pi \mathbf{1} = a$, while c is the current column marginal. Both P and Q are row stochastic.

Proposition 6 (Moving conditional metric). *At every finite state,*

$$J(y) = DF(y) = QP, \quad \text{diag}(c)J = \Pi^T \text{diag}(a)^{-1}\Pi. \quad (19)$$

Consequently J is self-adjoint and positive semidefinite in the c -weighted inner product, with spectrum in $[0, 1]$. For the actual displacement $d = F(y) - y$,

$$c = b \odot e^{-d}. \quad (20)$$

Proof. Differentiation gives $Dx[h] = -Ph$ and $DF[h] = QPh$. The second identity in (19) follows by multiplication. Conjugating J by $\text{diag}(\sqrt{c})$ gives $S^T S$, where $S = \text{diag}(a)^{-1/2} \Pi \text{diag}(c)^{-1/2}$. Jensen's inequality and the two marginal identities imply $\|Ph\|_a \leq \|h\|_c$ and $\|Qh\|_c \leq \|h\|_a$. The spectrum therefore lies in $[0, 1]$. Finally, $c = v \odot (K^T u)$ and $e^{F(y)} = b/(K^T u)$ imply (20). $\square$

For the remainder of this section the marginals have unit mass. The metric c is determined by the actual displacement, but the full operator J is not. Under a displacement h , put $e = -\log(Pe^h)$. Direct normalization gives

$$\begin{aligned} P^+ &= \text{diag}(Pe^h)^{-1}P \text{diag}(e^h), \\ Q^+ &= \text{diag}(Qe^e)^{-1}Q \text{diag}(e^e), \\ F(y+h) - F(y) &= -\log(Qe^e). \end{aligned} \quad (21)$$

These identities close the evolving geometry in a diagonal scaling orbit. They do not reduce its dense application cost. Nor does instantaneous self-adjointness imply that operators at successive anchors commute.

A. Curvature is controlled by conditional variance

Write $\Gamma_y(u, v) = D^2 F(y)[u, v]/2$ and define

$$\mathcal{E}_y(h) = \sum_i a_i \text{Var}_{P_i}(h) = \langle h, (I - J)h \rangle_c. \quad (22)$$

Let $\|h\|_{c,p}$ denote the probability-weighted L^p norm and $\text{osc}(h) = \max h - \min h$. The latter is the norm on log states modulo the constant gauge.

Theorem 7 (Conditional curvature and finite remainder). *At every finite anchor,*

$$\|\Gamma_y(u, v)\|_{c,1} \leq \sqrt{\mathcal{E}_y(u)\mathcal{E}_y(v)}. \quad (23)$$

For $R = \text{osc}(h)$, define $C_2(R) = 2\phi_2(2R)$ and $C_3(R) = 5R\phi_3(2R)$, where

$$\phi_k(s) = \frac{e^s - \sum_{j=0}^{k-1} s^j/j!}{s^k}, \quad \phi_k(0) = 1/k!.$$

Then

$$\|F(y+h) - F(y) - Jh\|_{c,1} \leq C_2(R)\mathcal{E}_y(h), \quad (24)$$

$$\|F(y+h) - F(y) - Jh - \Gamma_y(h, h)\|_{c,1} \leq C_3(R)\mathcal{E}_y(h). \quad (25)$$

These are finite-amplitude bounds at the given anchor; no fixed point or spectral separation is assumed.

Proof. Differentiating conditional means gives

$$2\Gamma_y(u, v) = Q \text{Cov}_P(u, v) - \text{Cov}_Q(Pu, Pv).$$

Conditional Cauchy-Schwarz bounds the first term in weighted L^1 by $\sqrt{\mathcal{E}_y(u)\mathcal{E}_y(v)}$. For the second, use

$$\sum_j c_j \text{Var}_{Q_j}(Pu) = \langle u, (J - J^2)u \rangle_c \leq \mathcal{E}_y(u),$$

and the same inequality for v . Division by two proves (23).

Along $y_s = y + sh$, exponential tilting gives $e^{-sR} \leq (P_s)_{ij}/P_{ij} \leq e^{sR}$ and the same comparison for $(c_s)_j/c_j$. Evaluating tilted variances about the old conditional mean yields $\mathcal{E}_{y_s}(h) \leq e^{sR}\mathcal{E}_y(h)$. Consequently $\|D^2 F(y_s)[h, h]\|_{c,1} \leq 2e^{2sR}\mathcal{E}_y(h)$.

With κ_3 the third central moment, a further derivative is

$$\begin{aligned} D^3 F(y)[h, h, h] &= Q\kappa_{3,P}(h) + \kappa_{3,Q}(Ph) \\ &\quad - 3\text{Cov}_Q(Ph, \text{Var}_P(h)). \end{aligned}$$

The bounds $|\kappa_3(X)| \leq \text{range}(X) \text{Var}(X)$ and $|\text{Cov}(X, V)| \leq \text{range}(X)\mathbb{E}V$ for $V \geq 0$ give $5R\mathcal{E}_y(h)$ in the current weighted L^1 norm. The two tilt comparisons give $5Re^{2sR}\mathcal{E}_y(h)$ in the anchor norm. Taylor's integral remainder proves both finite bounds. $\square$

For a normalized eigenvector $Jv = \lambda v$, $\mathcal{E}_y(v) = 1 - \lambda$. Thus slow directions have small quadratic forcing in the stated norms. This coefficientwise fact is not an analytic linearization radius or a guarantee that a single slow coordinate closes. Mixed forcing can still carry information from other directions.

B. A finite energy budget over arbitrary horizons

Positive linear maps and their log forms preserve order; the full F is order preserving and translation equivariant. Hence F is nonexpansive in oscillation. This supplies the common norm needed to accumulate defects.

Theorem 8 (Fixed-point finite-energy transport). *Assume $F(y_*) = y_*$, set $c = b$, $m = \min_j c_j > 0$, and let h be c -centered, fixing the constant gauge used to measure amplitude. Put $A = \|h\|_{c,2}$, $R = \text{osc}(h)$, $J = DF(y_*)$, and $\Gamma = \Gamma_{y_*}$. Define*

$$\begin{aligned} \ell_0 &= h, & q_0 &= 0, & \ell_{j+1} &= J\ell_j, \\ & & q_{j+1} &= Jq_j + \Gamma(\ell_j, \ell_j). \end{aligned}$$

For every finite H ,

$$\text{osc}\{F^H(y_* + h) - y_* - J^H h\} \leq C_2(R)A^2/m. \quad (26)$$

With $S = A^2/\sqrt{m}$ and $\bar{R} = R + A^2/m$,

$$\begin{aligned} &\text{osc}\{F^H(y_* + h) - y_* - \ell_H - q_H\} \\ &\leq \frac{2AS + S^2 + C_3(\bar{R})(A + S)^2}{m}. \end{aligned} \quad (27)$$

Uniformity is in H at fixed amplitude and positive marginal geometry. The bounds contain no inverse spectral gap; they do contain m^{-1} , R , and an unknown fixed-point displacement. For fixed c , the second bound is $O(A^3)$ as $A \rightarrow 0$.

Proof. Self-adjointness and $0 \leq J \leq I$ give

$$\sum_{j=0}^{H-1} \mathcal{E}(J^j h) \leq \|h\|_{c,2}^2 = A^2, \quad (28)$$

since $\mathcal{E}(w) \leq \|w\|_{c,2}^2 - \|Jw\|_{c,2}^2$. Also $\text{osc}(J^j h) \leq R$. Apply (24) to each linear-path defect, use $\text{osc}(w) \leq \|w\|_{c,1}/m$, and accumulate by nonexpansivity to obtain (26).

Set $f_j = \Gamma(\ell_j, \ell_j)$. Equations (23) and (28) imply $\sum_j \|f_j\|_{c,1} \leq A^2$, whence $\text{osc}(q_j) \leq A^2/m$. The impulse expansion of q_j , Minkowski's inequality in the space of energy sequences, and $\|f\|_{c,2} \leq \|f\|_{c,1}/\sqrt{m}$ give

$$\left(\sum_j \mathcal{E}(q_j) \right)^{1/2} \leq \sum_j \|f_j\|_{c,2} \leq S.$$

For $z_j = \ell_j + q_j$, therefore, $\text{osc}(z_j) \leq \bar{R}$ and $\sum_j \mathcal{E}(z_j) \leq (A + S)^2$. The quadratic discrepancy has total $L^1(c)$ norm at most $2AS + S^2$ by (23). The remaining Taylor defects sum to at most $C_3(\bar{R})(A + S)^2$. Converting to oscillation and using nonexpansivity proves (27). $\square$

This second jet is the linear evolution on $(z_j, \ell_j \otimes \ell_j)$ with block operator

$$\mathcal{L} = \begin{pmatrix} J & \Gamma_{\text{flat}} \\ 0 & J \otimes J \end{pmatrix}.$$

It retains resonant interactions. It is not repeated application of the quadratic Taylor map, whose higher-degree terms would evolve differently. The uncompressed tensor dimension is itself a substantial cost.

At a general anchor, a proposed path s_j instead has the defect

$$\eta_j = \text{osc}\{s_{j+1} - r - Js_j - \Gamma_y(s_j, s_j)\}, \quad r = F(y) - y.$$

The same argument gives a *pathwise* bound

$$\begin{aligned} &\text{osc}\{F^H(y + s_0) - y - s_H\} \\ &\leq \sum_{j < H} \left[\eta_j + \frac{C_3(\text{osc } s_j) \mathcal{E}_y(s_j)}{\min c} \right]. \end{aligned} \quad (29)$$

It is not a proved horizon-uniform extension of Theorem 8. If $s_j = CL^j \xi$, the energy sum can be composed by a Gramian:

$$\begin{aligned} G_H &= \sum_{j < H} (L^j)^T W L^j, & W &= C^T \text{diag}(c)(I - J)C, \\ P_{a+b} &= P_b P_a, & G_{a+b} &= G_a + P_a^T G_b P_a, & P_a &= L^a. \end{aligned}$$

Binary composition costs $O(r^3 \log H)$ for an r -dimensional dense lift. A path-radius bound and the model-defect sum are still needed. Recovering h from an observed residual through $(I - J)^{-1}$ can reintroduce inverse-gap sensitivity. The energy theorem removes a forcing obstruction, not the acquisition problem.

VI. WHAT MUST BE OBSERVED AND RETAINED

A narrow state trajectory need not determine its changing local rule. The distinction can be stated exactly for linearized output prediction and then tested without converting it into a universal impossibility claim.

Proposition 9 (Exact linear rule observability). *For $h^+ = Jh$ and required output Oh , the least dimension of an exact linear autonomous representation*

$$\xi = Rh, \quad \xi^+ = T\xi, \quad Oh = D_o \xi$$

valid for every $h \in \mathbb{R}^n$ is

$$r_* = \dim \text{span rows}\{O, OJ, \dots, OJ^{n-1}\}. \quad (30)$$

Proof. The identities $RJ = TR$, $O = D_o R$ force the row space of R to be J -invariant and to contain every displayed row. Conversely that row span is invariant by Cayley–Hamilton; a basis supplies R , T , and D_o . $\square$

To carry selected changing-rule entries, include in O the functionals $h \mapsto 2\langle v_i, \Gamma_y(h, v_j) \rangle_c$, not just the retained state coordinates. Thus adding rule outputs can increase r_* . Eliminating hidden variables also creates temporal memory: if $x^+ = Ax + Bz$, $z^+ = Cx + Dz$, then

$$x_{k+1} = Ax_k + BD^k z_0 + \sum_{j < k} BD^{k-1-j} Cx_j.$$

Delay coordinates may represent that information; omission does not make it disappear [14], [15].

A. A tolerance-dependent directional law

Let $T_H = [O^T, (OJ)^T, \dots, (OJ^{H-1})^T]^T$ and let the perturbation ensemble have second-moment matrix $\Sigma = \mathbb{E}[hh^T]$. In specified input and output coordinates put $M_H = \Sigma^{1/2} T_H^T T_H \Sigma^{1/2}$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$. If $\text{tr } M_H > 0$, the best rank- r linear stacked predictor has relative mean-square loss

$$1 - \frac{\sum_{i \leq r} \lambda_i}{\text{tr } M_H}. \quad (31)$$

This is the singular-value truncation theorem applied to $T_H \Sigma^{1/2}$. In particular its captured fraction is at most r/d_{eff} , where $d_{\text{eff}} = \text{tr } M_H / \lambda_1$. Capture η requires $r \geq \eta d_{\text{eff}}$ within this model class.

The law is exact for the specified linear prediction task and perturbation ensemble. It is a ratio of mean squared errors, not a worst-case relative bound. A 99% energy capture permits 10% relative RMS error. Moreover the optimal stacked predictor need not admit autonomous projected dynamics: a Gramian tail bound does not certify invariance or control nonlinear regeneration of discarded modes. Neither mixture count nor ambient dimension can replace this task-dependent spectrum without another theorem.

Finite observations alone also cannot certify unrestricted smooth future maps. A smooth perturbation supported away from the finitely queried states leaves every observed value and derivative unchanged and can change a later unqueried transition. Certification consequently needs a stated structural class, an invariant-region bound, or another source of information beyond snapshot rank. This does not forbid useful statistical prediction.

B. The algebra of nonlinear requests

There is a complementary way to avoid carrying an evolving Jacobian. Sinkhorn repeatedly requests Kv and $K^T u$, while its reciprocals remain cheap full-vector operations. The linear relation K is fixed even when its normalized conditional geometry changes. Its action on a retained request space can be measured once and reused.

Theorem 10 (Finite joint-pattern algebra). *Let $w_1, \dots, w_d \in \mathbb{R}^n$. Partition the indices by equality of the joint tuples $(w_1(i), \dots, w_d(i))$, and let N_{pat} be the number of nonempty classes. The unital pointwise algebra generated by the w_j is precisely the space of vectors constant on those classes; its dimension is N_{pat} . It is closed under componentwise exponentiation and reciprocals of vectors with no zero coordinate. If Q spans this algebra and $x_0 > 0$, measured columns of $K \text{diag}(x_0)Q$ evaluate every request*

$$K \left[x_0 \odot \exp \left(\sum_j s_j w_j \right) \right]$$

by linear combination.

Proof. Every generated product is constant on each class. Choose a linear combination $w = \sum_j \alpha_j w_j$ separating the finitely many distinct tuples; the forbidden coefficients form finitely many hyperplanes. Univariate Lagrange polynomials

in w give the indicator of each class. These indicators span all class-constant vectors, proving equality and the dimension claim. Any componentwise function defined on the finitely many values remains class-constant. Linearity of K proves the response claim. $\square$

For binary generators $N_{\text{pat}} \leq 2^d$; a single generator with n distinct values already has $N_{\text{pat}} = n$. Thus a small span of log directions need not produce a small exact algebra of requests. This is an obstruction to *exact linear-space closure on the full generated family*. It is not a lower bound for approximate representations, a restricted path, nonlinear parameterizations, or every acceleration method. It also does not prove that ordinary Sinkhorn remains in a chosen generator family. That membership must be established or checked at every actual request.

C. Admission and numerical transport of primitive responses

At a positive anchor x_0 let $k_0 = Kx_0$ and $P_0 = \text{diag}(k_0)^{-1} K \text{diag}(x_0)$. For an actual request $x > 0$, write $v = x/x_0 = Qc + r$ and $\hat{p} = P_0 Qc$. Positivity yields

$$\eta = \max_i |r_i|/v_i < 1 \implies |P_0 v - \hat{p}| \leq \eta P_0 v. \quad (32)$$

The represented output is positive and its Hilbert log-distance from the exact one is at most $\log((1+\eta)/(1-\eta))$. Positive linear maps are nonexpansive in that distance, while reciprocals and fixed positive diagonal scalings preserve it. Summing the admitted half-pass errors bounds departure from the original ordinary trajectory.

This gate uses the full observed input, not a prediction of what the next request ought to be. If each stored response has a proved componentwise error matrix E , the additional output uncertainty is bounded by $E|c|$; otherwise (32) covers exact arithmetic only. The numerical results use double precision with roundoff allowances and independent comparisons against direct evaluation. These checks do not constitute directed-rounding interval certification.

At another anchor x_1 with known $k_1 = Kx_1$, measured pairs transform as

$$\begin{aligned} P'_0 &= \text{diag}(k_0/k_1) P_0 \text{diag}(x_1/x_0), \\ Q' &= \text{diag}(x_0/x_1) Q, \quad P'_0 Q' = \text{diag}(k_0/k_1) P_0 Q. \end{aligned}$$

This is an exact change of representation. Its input diagonal condition number is $\exp(\text{osc } \log(x_1/x_0))$; near-dependent basis transformations can further amplify response errors. Stable frame transfer, useful future request coverage, and cheap acquisition are therefore different questions. Exact algebra settles none of their numerical cost comparisons by itself.

VII. EMPIRICAL CONSEQUENCES OF THE THEORY

The following outcomes are selected because they distinguish closure, acquisition, and cost. They are not a pooled benchmark: a finite-trajectory comparison, a complete solve, and a primitive-query count answer different questions. Table I records the strongest positive examples alongside the relevant limitations. Paired timings use the same computing environment and include method-specific setup and validation; Appendix F specifies the individual comparison contracts.

TABLE I
SELECTED OUTCOMES ORGANIZED BY THE CONDITION THEY TEST. RATIOS COMPARE AGAINST THE STATED ORDINARY IMPLEMENTATION, NOT AGAINST ALL AVAILABLE SOLVERS.

Representation	Source of validity	Reported outcome and its scope
Huber affine piece	Current-state discovery; feature-path certificate	At $n = 256$ with positive sensing, complete solves were $3.47\times$ faster than ordinary and $1.42\times$ faster than the fixed schedule. Anderson was faster still. Signed sensing made discovery unprofitable.
Two-group Sinkhorn	Supplied exact proportional-block partition; projective powers	$4.7\text{--}6.9\times$ faster at coupling .03 and $56.9\text{--}73.8\times$ at .001, including entry/readout against the same reduced map. Partition discovery is not measured.
Primitive response cache	Measured fixed-kernel actions; full-request residual admission	At $n = 1024$, four of six 256-pass trajectories ran $1.13\text{--}2.04\times$ faster. The two narrowest-kernel cases and all $n = 128$ cases lost time.
Coupled Meyer path	Exact future-driving quotient; curved remainder certificate	Seven of twenty anchors were accepted: four ran $1.39\text{--}3.48\times$ faster and three lost time. The discovered policy lost all five complete solve comparisons.
Mixed request algebra	Joint-pattern closure or checked approximate membership	Four binary directions used 16 measured responses for 256 supplied-family requests. Four Gaussian directions at budget 64 obtained no reuse. Small- n product savings did not give elapsed-time savings.

A. Where a relation becomes an executable shortcut

The Huber example acquires the drift, its relation, and the first feature event from the current state. At $n = 256$, three positive-sensing instances admitted initial exact-arithmetic horizons 838, 831, and 821 with one map and one tangent action; independent replay errors were at most $3.7 \cdot 10^{-11}$ in absolute whitened dual norm. The complete-solve speedup in Table I is a separate measurement, including later failed acquisitions and checks. At smaller size or with frequent feature events, those costs were not repaid.

The proportional-block example instead begins with the invariant family supplied. Both methods aggregate the same inputs, evolve the same reduced map, and reconstruct the full scalings. For coupling .001, ordinary reduced runs took about 14–15 ms while projective powers and readout took .19–.26 ms. This isolates economical composition after entry; it cannot be used as evidence that arbitrary kernels reveal such a partition.

The rational cross-ratio control occupies an intermediate position. Eight transitions supply observations in known mirror coordinates; a finite rational grammar identifies the chart. Across four mirror geometries and three seeds, the maximum relative primal error at horizon 128 was $2.88 \cdot 10^{-11}$. This verifies a favorable representation-discovery mechanism, not a general grammar or a repeated wall-clock performance gate.

B. Small forcing does not establish cheap acquisition

The conditional-energy study checked 50 finite remainder bounds and 28 fixed-point horizon/amplitude cases. All finite bounds held in those floating-point checks. As a controlled spectral gap decreased from .386 to .00145, the horizon $\lceil 3/\text{gap} \rceil$ grew from 8 to 2070. At amplitude .1, maximum second-jet error changed from $1.13 \cdot 10^{-4}$ to $1.17 \cdot 10^{-4}$. At the smallest gap, halving amplitude twice reduced errors to

$1.46 \cdot 10^{-5}$ and $1.82 \cdot 10^{-6}$, consistent with the cubic bound rather than an inverse-gap blowup.

The bound itself can nevertheless be unusable. Replicating every atom 32 times, from three states to 96, preserves the physical conditional action but worsens the coarse marginal-dependent certificate by approximately 1600-fold. The experiment diagnoses the $\min c$ conversion; it does not provide a proved discretization-independent replacement. Likewise, the fixed-point displacement and full second jet are scoring objects in this study, not free runtime information.

The observability tests expose a separate limitation. At $n = 128$, the four-by-four reference rule block required 12–23 directions to retain 99.9% of its frozen observability-Gramian trace, whereas its four state coordinates alone require four. These are trace-capture ranks, not worst-case relative-error certificates. At early anchors of the two $\varepsilon = .001$ cases, more than 32 and 28 current leading modes were needed to capture 99% of the fixed point’s leading direction; both late anchors needed one. This is evidence against a uniformly tiny leading cluster over that acquisition window, not against every other representation.

History was not categorically uninformative. An order-eight recurrence, given a favorable oracle signal of frozen rule drift, predicted 32 held-out values to within 3% relative RMS error in 21 of 24 cases with relative noise 10^{-6} . The fixed point and exact modal signal were supplied to this diagnostic. It sets a target for causal acquisition; it is not an implemented general acquisition guarantee.

C. Pattern closure and its loss under richer requests

The joint-pattern control uses 256 positive requests on 128 coordinates, two seeds, and a 64-column budget. For supplied binary fields with one, two, and four directions, acquisition and all queries cost 2, 4, and 16 kernel-vector equivalents. Learning directions from the first eight live log requests costs

9, 11, and 23–27 equivalents. All subsequent requests were admitted. Separate powers of the original directions fail in the multidirection cases, demonstrating the role of mixed products.

With continuous Gaussian fields, supplied one-direction representations still admitted all subsequent requests. Two directions cost 279 and 266 products, exceeding the ordinary 256; four cost 319 in both seeds and admitted none. Total log-coefficient RMS was held fixed; maximum log range was not. These results do not establish a universal law in Gaussianity. The precise exact-family statement is Theorem 10; approximate cost additionally depends on range, tolerance, conditioning, and the portion of the family actually visited.

On the difficult ordinary Sinkhorn trajectories at $n = 1024$, $\epsilon = .001$, two learned directions required 1016 products per seed for 256 passes, against the ordinary 512. Four directions required 1015 and 1012, admitting only one and four reuses. Acquisition consumed nearly all the extra work. After rejection, the already-required full product audited the rejected prediction: none of the 496 rejected predictions per seed in the two-direction case had componentwise relative output error below 10^{-8} . Only two of 495 and one of 492 did so with four directions. A looser gate would therefore not repair the dominant representation failure. These diagnostics do not isolate generator insufficiency from polynomial truncation.

Frame conditioning is distinct again. Transferring measured responses through repeated ill-conditioned basis changes initially corrupted their images. Carrying response uncertainty, spectral rebasing, and immutable physical action pairs repaired the tested accuracy but still lost complete cost. The algebraic frame identity does not guarantee either numerical stability or future request coverage.

VIII. CONDITIONS FOR PLAUSIBLE ACCELERATION

A transport representation is promising when the required outputs close in a space that can be acquired before its validity expires, and when its composition and validation are cheaper than the work replaced. The theory gives several concrete ways this can happen: a discovered affine piece with a distant event, a supplied finite invariant family with cheap entry, a controlled finite-energy path with an economical certificate, or a sequence of actual requests well covered by measured primitive responses. These mechanisms carry different objects and need different evidence.

The corresponding failures are equally distinct. A small frozen space may miss changing geometry; a geometrically narrow path may hide rule memory; a small set of log directions may generate a large multiplicative family; and a mathematically exact frame may be numerically expensive or unstable. Equation (31) provides a sharp directional law for a specified linear output problem. Theorem 10 supplies an exact dimension law for a specified pointwise algebra. Neither establishes an intrinsic scalar complexity law for arbitrary mixture models or all forms of acceleration.

The finite-energy theorem explains why slow relaxation alone need not defeat approximate transport: its weak dissipation also limits accumulated nonlinear forcing. The unresolved step is obtaining and advancing the required information economically at accessible anchors. Small Gramian tails, accurate

local fits, and valid but conservative bounds each address only part of that step. A practical theory must retain their acquisition and conditioning constants alongside their approximation rates.

Discoverable transport is therefore a question about relations preserved by computation. Exact families show that substantial shortcuts are possible; scoped obstructions show why the same representation need not extend to a richer family. The empirical outcomes locate examples on both sides without turning a failed carrier into a general impossibility theorem, or a successful supplied family into a universal discovery procedure.

APPENDIX A

QUADRATIC FINITE FLOW AND ITS CLASSICAL CONTENT

This section identifies precisely the classical linear-algebraic mechanism. Let $\psi(x) = \frac{1}{2}x^T G x$, $G \succ 0$, and $f(x) - f(x_*) = \frac{1}{2}(x - x_*)^T B(x - x_*)$, $B \succ 0$. Then $y = Gx$, $M = G^{-1}$, and $J = I - \alpha B M$ are constant. Assume the eigenvalues of $G^{-1/2} B G^{-1/2}$ lie in $[\mu, L]$ and $0 < \alpha \leq 1/L$. Set

$$q = 1 - \alpha\mu, \quad \kappa = L/\mu, \quad \theta = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}, \quad (33)$$

$$W = M(I - J) = \alpha M B M.$$

The squared W -norm of the dual error equals 2α times the objective gap.

Theorem 11 (Finite-horizon Galerkin contraction). *In exact arithmetic, the natural-metric proposal satisfies*

$$\frac{f(\hat{x}_H) - f(x_*)}{f(x) - f(x_*)} \leq q^{2H} + (1 - q^{2H}) \min\{1, 4\theta^{2k}\}, \quad (34)$$

where $\hat{x}_H = G^{-1}\hat{z}_H$. More precisely, define the stationary Galerkin correction

$$\delta_\infty = \beta Q(I - T_r)^{-1} e_1, \quad e_\infty = y + \delta_\infty - y_*.$$

Then the finite-horizon error has the orthogonal decomposition

$$\|y + \delta_{k,H} - y_*\|_W^2 = \|e_\infty\|_W^2 + \|\delta_{k,H} - \delta_\infty\|_W^2. \quad (35)$$

Proof. Write $S = I - J$, $e = y - y_*$. Since $r = -Se$ and $\mathcal{K}_k(J, r) = \mathcal{K}_k(S, r)$, δ_∞ is the M -Galerkin solution of $S\delta = r$. Thus $Q^T W e_\infty = 0$. Both $\delta_{k,H}$ and δ_∞ lie in $\text{ran } Q$, proving (35). The reduced matrix T_r is symmetric with spectrum in $[0, q]$, while $Q^T W Q = I - T_r$. Consequently

$$\delta_\infty - \delta_{k,H} = \beta Q T_r^H (I - T_r)^{-1} e_1, \\ \|\delta_\infty - \delta_{k,H}\|_W \leq q^H \|\delta_\infty\|_W.$$

Galerkin orthogonality also gives $\|e\|_W^2 = \|e_\infty\|_W^2 + \|\delta_\infty\|_W^2$. Combine these identities with the classical CG polynomial bound [5], $\|e_\infty\|_W \leq 2\theta^k \|e\|_W$, and with its trivial bound $\|e_\infty\|_W \leq \|e\|_W$. Converting squared energy to objective gap proves (34). $\square$

This theorem does not establish a competitor to CG on a fixed quadratic. The stationary Galerkin correction has no larger energy error than its finite-horizon truncation. The finite polynomial becomes interesting when the geometry changes, a finite trajectory is itself the target, or a stationary local equation is ill-conditioned or inconsistent. On an affine quadratic, classical CG is a necessary reference point.

APPENDIX B

AFFINE-PIECE DISCOVERY AND EVENT HORIZONS

A class with stronger available structure permits actual certification. Consider the Huber inverse problem

$$f(x) = \sum_i h_\delta((Ax - b)_i), \quad \psi(x) = \frac{1}{2}x^T Gx, \quad G \succ 0,$$

where $h'_\delta(a) = \text{clip}(a, -\delta, \delta)$. In whitened dual coordinates $t = G^{1/2}x$, put $B = AG^{-1/2}$. The map is

$$\mathcal{F}(t) = t - \alpha B^T \text{clip}(Bt - b, -\delta, \delta), \quad 0 < \alpha \leq 2/\|B\|_2^2. \quad (36)$$

It is globally nonexpansive in Euclidean t -norm: the difference of two updates is $(I - \alpha \bar{B})(t - s)$, where the averaged Hessian obeys $0 \preceq \bar{B} \preceq \|B\|_2^2 I$. Hence the eigenvalues of the difference operator lie in $[-1, 1]$. Fix an anchor t , residual $a = Bt - b$, and mask $D_{ii} = 1_{\{|a_i| < \delta\}}$. On the corresponding closed affine piece,

$$J = I - \alpha B^T D B, \quad \mathcal{F}(t + v) = \mathcal{F}(t) + Jv.$$

The projected path is $\hat{t}_j = t + Qc_j$, where $c_0 = 0$ and $c_{j+1} = \beta e_1 + T_r c_j$. Every piece constraint can be checked using only

$$a_j = a + (BQ)c_j. \quad (37)$$

The columns Bq_i are already computed by the tangent actions and cached. The checks require feature-space operations, not another B^T application or another full map. Their execution time is nevertheless charged.

Theorem 12 (An affine-piece discovery certificate). *If (37) remains in the anchor's closed piece for every input index $j = 0, \dots, H-1$, and $E = JQ - QT_r$, then*

$$\|\mathcal{F}^H(t) - (t + Qc_H)\|_2 \leq \sum_{j=0}^{H-1} \|Ec_j\|_2. \quad (38)$$

In particular, algebraic closure $E = 0$ makes the discovered finite transport exact, even when its final output crosses a piece boundary.

Proof. The checked affine formula gives $\mathcal{F}(\hat{t}_j) - \hat{t}_{j+1} = Ec_j$ at every predicted input. For the actual state t_j , nonexpansivity gives $\|t_{j+1} - \hat{t}_{j+1}\| \leq \|t_j - \hat{t}_j\| + \|Ec_j\|$. Induction proves the bound. It is not necessary to assume that the unknown actual path stays inside the same piece. The last checked point is input $H-1$, not output H . $\square$

The norms in (38) use the small Gram matrix $E^T E$. This gives a computable path certificate in exact arithmetic and a floating-point diagnostic subject to a stated numerical tolerance; we do not claim directed-rounding certification. An exterior Euclidean disk projection is curved, so this theorem does not transfer to Meyer merely because its inside/outside mask stays unchanged.

A. Unit-mode drift and the Anderson distinction

If all Huber residuals are saturated, then $D = 0$, $J = I$, and $r = -\alpha B^T \text{sign}(a)\delta$ is constant until the next event. The first Arnoldi action discovers a rank-one cyclic space, and $p_H(J)r = mr$. Put $d = Br$. For every feature moving toward its nearest boundary, the continuous crossing time is

$$\tau_i = \frac{\text{sign}(a_i)\delta - a_i}{d_i} > 0.$$

The horizon $H = 1 + \lfloor \min_i \tau_i \rfloor$ has all inputs in the current closed piece and may place its output beyond the first boundary. A finite cap limits acquisition amortization. In finite precision, each proposed input must also be checked against its defining feature inequalities; an uncertain boundary point cannot establish exact piece membership. Thus the drift direction, one-dimensional relation, and first-event horizon are all obtained from the current problem. They are not supplied by a known optimum or by a future reference trajectory.

For type-II Anderson using a history entirely within this same constant-drift piece, every residual difference is zero. With the minimum-norm or positive ridge least-squares convention, its extrapolation coefficients are zero: it takes one ordinary step. This is a precise collapse of acceleration, not divergence of the underlying iteration. More generally, a left invariant direction $w^T J = w^T$ with $w^T r \neq 0$ obstructs a stationary correction because $w^T(I - J)\delta = 0$, while $w^T p_H(J)r = mw^T r$ retains the required drift. A locally persistent pattern can therefore carry an unresolved future event. History spanning other pieces, roundoff, or alternative regularization can change the behavior; no universal Anderson failure is asserted. The statement concerns this specified history and regularization convention.

APPENDIX C

CURVED VALIDITY IN THE COUPLED MEYER MAP

The affine-piece certificate does not apply to an exterior disk projection. Here we derive a different certificate from the original coupled recurrence, retaining its curvature and both vector memories. The resulting construction requires no additional screened solves during path validation.

A. A faithful future-driving quotient and its metric

Let D be the periodic discrete gradient and $D^* = -\text{div}$. For a branch with $a = \eta/c > 0$, define

$$S_a = (I + aD^*D)^{-1}, \quad R_r(t) = t - 2\Pi_r(t),$$

where Π_r is the pointwise Euclidean disk projection. The exact stored Meyer state is (u, w, t_u, t_w) . Its next pass depends on the primals only through $s = u + w$:

$$\begin{aligned} u^+ &= S_a\{s + aD^*R_{r_u}(t_u)\}, \\ w^+ &= S_b\{f - u^+ + bD^*R_{r_w}(t_w)\}, \\ t_u^+ &= Du^+ + \Pi_{r_u}(t_u), \\ t_w^+ &= Dw^+ + \Pi_{r_w}(t_w), \quad s^+ = u^+ + w^+. \end{aligned} \quad (39)$$

The reported experiment sets $a = 2$ and $b = 10$. Thus $z = (s, t_u, t_w)$ is an exact state for all future passes. Both vector

fields, including their transverse components, remain present. The current primal difference $u - w$ is not recoverable from this quotient; the next actual pass recovers both next primals exactly. The emitted texture $f - u - w = f - s$ is already recoverable at the current state. This is a precise quotient of redundant state, not deletion of a driving memory variable.

Theorem 13 (Metric of the coupled recurrence). *The quotient map F in (39) is nonexpansive in*

$$\|z\|_M^2 = \|s\|_2^2 + a\|t_u\|_2^2 + b\|t_w\|_2^2. \quad (40)$$

In fact, for any two states $z, \tilde{z}$,

$$\|Fz - F\tilde{z}\|_M^2 + \frac{1}{2}\|(I - F)z - (I - F)\tilde{z}\|_M^2 \leq \|z - \tilde{z}\|_M^2. \quad (41)$$

Equivalently, this map is 2/3-averaged. The statement does not require the input states to lie on an ordinary proximal trajectory.

Proof. On pairs (y, t) with norm $\|y\|^2 + a\|t\|^2$, let $P_{\mathcal{G}_a}$ be the orthogonal projection onto the graph $\mathcal{G}_a = \{(x, Dx)\}$. Its first component on $(y, R_r(t))$ is $v = S_a(y + aD^*R_r(t))$. Therefore the branch map

$$\begin{aligned} A_a(y, t) &= (v, Dv + \Pi_r(t)) \\ &= \frac{1}{2}\{I + (2P_{\mathcal{G}_a} - I)\text{diag}(I, R_r)\}(y, t) \end{aligned}$$

is firmly nonexpansive: graph reflection is an isometry, and R_r is nonexpansive by firm nonexpansivity of disk projection.

Write differences between two inputs and outputs with a Δ ; in the following inequalities, u, w denote the updated primals. The first branch gives

$$\begin{aligned} \|\Delta u\|^2 + a\|\Delta t_u^+\|^2 \\ \leq \|\Delta s\|^2 + a\|\Delta t_u\|^2 \\ - \|\Delta(s - u)\|^2 - a\|\Delta(t_u - t_u^+)\|^2. \end{aligned}$$

Firm nonexpansivity of the second branch, whose input difference is $(-\Delta u, \Delta t_w)$, gives

$$\|\Delta w\|^2 + b\|\Delta t_w^+\|^2 \leq -\langle \Delta w, \Delta u \rangle + b\langle \Delta t_w^+, \Delta t_w \rangle.$$

Rearranging twice this inequality shows

$$\begin{aligned} \|\Delta(u + w)\|^2 + b\|\Delta t_w^+\|^2 \\ \leq \|\Delta u\|^2 + b\|\Delta t_w\|^2 \\ - \|\Delta w\|^2 - b\|\Delta(t_w - t_w^+)\|^2. \end{aligned}$$

Combining the two bounds proves nonexpansivity. Finally,

$$\|\Delta(s - u)\|^2 + \|\Delta w\|^2 \geq \frac{1}{2}\|\Delta(s - u - w)\|^2$$

yields (41) with the two vector-memory terms. $\square$

Whitening this metric gives ordinary Euclidean Arnoldi on the five scalar fields $(s, \sqrt{a}t_u, \sqrt{b}t_w)$. This five-field metric and a six-field Euclidean metric generally give different projected polynomials. Exact state equivalence does not make their compressed approximations identical.

B. Exact algebraic description of disk curvature

For an anchor vector $t = \rho n$, choose an orthonormal frame (n, e) and write a reduced displacement as $h = h_n n + h_t e$. Both h_n and h_t are linear forms in the Krylov coordinates c . The new squared radius is

$$\ell(c)^2 = (\rho + h_n(c))^2 + h_t(c)^2. \quad (42)$$

It is a quadratic polynomial. The disk event is the inequality $\ell(c)^2 \leq r^2$; normalization introduces its positive square root. These relations can be acquired once from the anchor and basis.

Let $q = \Pi_r(t + h) - \Pi_r(t) - P_t h$, where P_t is the selected linear projection action used by Arnoldi. For an interior anchor, including the chosen identity action at $\rho = r$,

$$\|q\|^2 = (\ell - r)_+^2. \quad (43)$$

For an exterior anchor $\rho > r$, put $\sigma = \min(1, r/\ell)$, with $\sigma = 1$ at $\ell = 0$. Then

$$q_n = \sigma(\rho + h_n) - r, \quad q_t = (\sigma - r/\rho)h_t. \quad (44)$$

These are exact identities on both sides of a crossing. In particular, pure radial motion on the same exterior ray has zero remainder while turning generally does not. A constant inside/outside mask alone fails to capture this distinction. At nonsmooth boundary anchors, the argument uses the specified linear action and exact remainder, not a nonexistent classical Jacobian.

Only two linear forms per disk site are needed to evaluate (42)–(44). The proposed representation is therefore a finite collection of polynomial radicands, branch inequalities, and radial normalization relations. It does not claim global finite closure of the nonlinear iteration in a polynomial observable dictionary.

C. The nonlinear remainder has a fixed linear response

Let q_u, q_w be the exact projection remainders at a candidate displacement. The corresponding remainder of the complete map has primal components

$$\begin{aligned} d_u &= -2aS_a D^* q_u, \\ d_w &= -S_b d_u - 2bS_b D^* q_w, \end{aligned} \quad (45)$$

and quotient components $(d_u + d_w, Dd_u + q_u, Dd_w + q_w)$. Thus all nonlinear curvature is confined to the two pointwise projection remainders; the rest is a fixed linear response. We do not evaluate the screened solves in (45) for each predicted point. Instead, their operator norm supplies a bound.

For a discrete Laplacian eigenvalue λ , longitudinal components of this response, after whitening, have Gram matrix

$$\begin{aligned} G(\lambda) &= \begin{pmatrix} 1 - d(\lambda) & g(\lambda) \\ g(\lambda) & 1 \end{pmatrix}, \\ d(\lambda) &= \frac{4a\lambda}{(1 + a\lambda)^2(1 + b\lambda)}, \\ g(\lambda) &= \frac{2\sqrt{ab}\lambda}{(1 + a\lambda)(1 + b\lambda)}. \end{aligned}$$

Transverse components have gain one. Consequently, with

$$\gamma^2 = \max_{\lambda \in \text{spec}(D^*D)} \left[1 - \frac{d(\lambda)}{2} + \sqrt{\frac{d(\lambda)^2}{4} + g(\lambda)^2} \right], \quad (46)$$

we have

$$\|N(v)\|_M \leq \gamma \sqrt{a \|q_u(v)\|_2^2 + b \|q_w(v)\|_2^2}. \quad (47)$$

This is a grid- and parameter-dependent calculation, independent of the source image. For the reported 64×64 grid, $\gamma = 1.14819$. The Gram formula follows by Fourier diagonalization of (45); the two transverse components pass unchanged. Independent tests compare the full nonlinear remainder against the linear response and its norm bound.

Theorem 14 (Curved Meyer path certificate). *For an M -orthonormal Arnoldi basis, let $E = JQ - QT_r$, $c_0 = 0$, $c_{j+1} = \beta e_1 + T_r c_j$, and $\hat{z}_j = z + Qc_j$. Define*

$$\varepsilon_j = \|Ec_j\|_M + \gamma \sqrt{a \|q_u(Qc_j)\|_2^2 + b \|q_w(Qc_j)\|_2^2}.$$

Then, for every finite horizon,

$$\|F^H(z) - \hat{z}_H\|_M \leq \sum_{j=0}^{H-1} \varepsilon_j. \quad (48)$$

The same bound controls the emitted texture error. It also remains valid after a common ordinary settling step. No fixed-mask assumption is required.

Proof. The exact defect is $N(Qc_j) + Ec_j$. Apply (47) and the triangle inequality, then propagate errors by Theorem 13. The initial error is zero, so summation gives (48). Texture is $f - s$ and its error norm is bounded by the full quotient norm. A common settling step cannot increase the quotient error. $\square$

All projection remainders along the predicted path are evaluated; this is not sparse sampling of a few convenient points. The reduction is algebraic: small-coordinate evaluation and pointwise radial operations replace repeated screened solves. Compression norms use the small matrix $E^T M E$. The certificate is proved in exact arithmetic and tested in floating point; we do not claim outward-rounded numerical certification.

APPENDIX D PROJECTIVE SINKHORN CLOSURE

Proposition 15 (Two-by-two Sinkhorn lift). *For $K = \begin{pmatrix} p & q \\ r & s \end{pmatrix} > 0$, the quotient $t = v_1/v_2$ obeys $t^+ = (At + B)/(Ct + D)$, where*

$$\begin{aligned} A &= (b_1/b_2)(qa_1r + sa_2p), & B &= (b_1/b_2)qs(a_1 + a_2), \\ C &= pr(a_1 + a_2), & D &= pa_1s + ra_2q. \end{aligned}$$

With $L = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$,

$$\begin{aligned} t_H &= \frac{(L^H)_{11}t_0 + (L^H)_{12}}{(L^H)_{21}t_0 + (L^H)_{22}}, \\ \frac{dt_H}{dt_0} &= \frac{\det(L)^H}{\{(L^H)_{21}t_0 + (L^H)_{22}\}^2}. \end{aligned} \quad (49)$$

If $ps \neq qr$, the one-pass derivative $\det(L)/(Ct + D)^2$ is nonconstant in t .

Proof. In any scale gauge, insert $u_1 = a_1/(pt + q)$ and $u_2 = a_2/(rt + s)$ into $t^+ = (b_1/b_2)(qu_1 + su_2)/(pu_1 + ru_2)$ and clear denominators. Composition of fractional-linear maps multiplies their homogeneous matrices. Differentiating the resulting fraction proves the second formula; $\det(L) = (b_1/b_2)a_1a_2(ps - qr)^2$. $\square$

Cayley–Hamilton expresses $L^H = \alpha_H L + \beta_H I$, and normalized binary powering needs $O(\log H)$ fixed-size matrix products. This transports a changing physical derivative using a constant lifted operator. Exact rational bit lengths and floating-point conditioning still require attention. The lift describes quotient states. To recover the ordinary post-column coupling after H passes, advance through $H - 1$ and perform the final row/column pair in a consistent gauge; a row-normalized output is obtained by evaluating the next row normalization.

The closure extends to arbitrary dimensions with supplied exact structure

$$K_{ij} = r_i C_{g(i),h(j)} s_j. \quad (50)$$

Let $A_g = \sum_{i:g(i)=g} a_i$, $B_h = \sum_{j:h(j)=h} b_j$. After one full pass, $v_j = (b_j/s_j)w_{h(j)}/B_{h(j)}$, and substitution gives the invariant reduced map

$$w^+ = B/\{C^T(A/(Cw))\}. \quad (51)$$

For arbitrary initial v , use $w_h = \sum_{j:h(j)=h} s_j v_j$ for the entry pass. Two groups on each side reduce exactly to Proposition 15. Within-manifold derivative evolution follows the same lift; arbitrary initial perturbations also require the derivative of the entry map. Discovery or verification of the block structure must be charged, and rank two alone does not imply (50).

APPENDIX E GAUSSIAN PRECISION CONVENTION

Take the continuous Lebesgue kernel $K(x, y) = \exp(-\|x - y\|^2/(2t))$ and centered Gaussian marginal densities with covariance matrices $A, B \succ 0$. If the column scaling has quadratic exponent $-y^T V y/2$, write $p = t^{-1}I + V \succ 0$. Completing the square in Kv gives a row exponent whose precision is $t^{-1}I - t^{-2}p^{-1}$. Dividing the source density by that integral therefore gives row-scaling precision

$$U = A^{-1} - t^{-1}I + t^{-2}p^{-1}.$$

The next column integral has conditional precision $t^{-1}I + U = A^{-1} + t^{-2}p^{-1} \succ 0$. Dividing the target density by it yields

$$\begin{aligned} p^+ &= B^{-1} + t^{-2}(A^{-1} + t^{-2}p^{-1})^{-1} \\ &= B^{-1} + (t^2 A^{-1} + p^{-1})^{-1}, \end{aligned}$$

which proves (7). The scaling functions themselves need not be integrable separately; the stated conditional integrals are. Scalar normalization constants do not affect this covariance recurrence.

The reported two-dimensional controls include isotropic, commuting anisotropic, and rotated covariance pairs at four regularization values. The isotropic $t = .001$ control agrees

with ordinary precision iteration at horizon 2048 to approximately $3.4 \cdot 10^{-14}$ relative error. Generic block powering is not uniformly stable in the anisotropic controls: ill-conditioned projective denominators can spoil finite-precision readout. Exact Riccati closure is therefore a structural example, not a claim that this generic implementation is a robust accelerator for every covariance.

APPENDIX F EXPERIMENTAL COMPARISON CONDITIONS

Each experiment uses its own matched ordinary control; kernels and problem setup are shared or excluded consistently within that experiment. Future ordinary trajectories, dense spectra, and converged fixed points are used only for scoring unless explicitly labeled as supplied oracle information. Analytic certificates are exact-arithmetic results checked in float64; no interval-arithmetic implementation is claimed.

A. Discovery and supplied structure

The Huber complete-solve target is objective-gap ratio 10^{-3} , checked every sixteen work units; the constructed noiseless inverse problem has known zero optimum. Three seeds and three shuffled repeats include failed acquisitions, checks, and fallback steps. At $n = 256$, the 3.47 ratio is the median of paired per-seed speed ratios. The Anderson implementation is an ungarded regularized type-II control, not a faithful reproduction of AA-BPG. Its better complete time does not certify reproduction of the ordinary intermediate trajectory.

The supplied-block Sinkhorn study uses sizes 256–8192 and coupling values .03, .001, with both methods operating on the same reduced map. Full-scaling readout is included. Its stopping test is maximum relative column error 10^{-9} after exact row normalization. Five repeats after warm-up are used; the reported tables contain 120 structured solves. The stated 56.9–73.8 range concerns only the .001 family.

The Meyer coverage study uses five 64×64 sources and four ordinary anchor prefixes, with depths 2, 4, 8 and horizons 8, 16, 32, 64. All 240 path comparisons satisfy the bound within numerical tolerance. The live policy accepts seven of twenty anchors; three accepted blocks still lose time. Complete solves use the same recovered primal–dual gap, target 10^{-4} of its initial value, and checks every 32 work units. Discovery and all rejections are charged. All five complete solves finish; all five discovered runs lose against ordinary iteration.

B. Primitive trajectories and family controls

The discrete transport instances use independent point clouds $X_i, Y_j \in [0, 1]^2$ sampled uniformly. Set $C_{ij} = \|X_i - Y_j\|_2^2 / \max_{k,l} \|X_k - Y_l\|_2^2$ and $K_{ij} = \exp(-C_{ij}/\varepsilon)$. The source and target masses are separately normalized samples of $w(z) = .05 + \sum_{\ell} \exp(-\|z - c_{\ell}\|_2^2 / .02)$, with two independently uniform centers for the source and three for the target. The six cache comparisons combine two independent instances with $\varepsilon \in \{.01, .003, .001\}$; all start from unit column scaling and allow at most 32 stored response directions. Primitive-cache comparisons use 256 complete passes from

the same initial scaling, with three repeats and single-threaded dense linear algebra. The per-query exact-arithmetic admission threshold is 10^{-8} . At $n = 1024$ the six final Hilbert log-state errors are $1.37 \cdot 10^{-8}$ – $3.18 \cdot 10^{-8}$; these are errors relative to the ordinary finite trajectory, not marginal residuals or objective gaps. The four speedups and two regressions in Table I refer to these six matched finite trajectories.

The mixed-algebra family control fixes the log-coefficient RMS scale at 1.4 and uses one, two, or four standardized fields, binary or Gaussian. Supplied generators are distinguished from SVD generators learned after eight live requests. The response budget is 64 measured columns. Maximum observed Hilbert output error is $1.13 \cdot 10^{-8}$ and maximum observed bound excess is $9.46 \cdot 10^{-14}$. At $n = 128$ the mixed methods take approximately 6–13 ms against .6 ms ordinary evaluation despite favorable binary product counts.

The difficult-trajectory comparison uses the same random point and mass construction at $n = 1024$, regularization .001, two independent instances, a budget of 64 response columns, and three timed repeats. Its median ordinary times are 73 and 61 ms; two-direction mixed times are 216 and 217 ms, with retrospective rejection diagnostics included. The product-count failure does not depend on those diagnostic costs. The largest new-carrier terminal Hilbert error is $2.13 \cdot 10^{-10}$; most requests are evaluated by ordinary fallback.

C. Proof checks and scope of the evidence

Independent numerical checks cover mirror symmetry, affine Arnoldi reproduction, the nonlinear defect identity, quadratic Galerkin contraction, Huber feature-event admission, Meyer quotient and curvature identities, conditional derivatives, finite energy sums, observability Gramians, Gaussian precision recurrences, and joint-pattern algebra closure. Exact-rational checks are used where available; empirical comparisons are not substituted for the proofs stated in the text.

The acquisition diagnostic covers 14 cases and 42 causal anchors, using dense fixed-point geometry solely for scoring. Its Gramian trace ranks and favorable history experiment are not runtime guarantees. The selected measurements distinguish the theoretical conditions they test; they do not establish a universal performance ordering.

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